

AI-driven fast-food restaurants: how perceived service quality, perceived smart quality and experiential value shape habit, satisfaction and customer loyalty

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Abstract

Purpose – This study aims to examine how perceived service quality and perceived smart quality jointly influence customer loyalty in AI-driven fast-food restaurants in China, with perceived experiential value, satisfaction and habit as mediating mechanisms.

Design/methodology/approach – The study draws on the stimulus–organism–response (SOR) framework, SERVQUAL and habit theory. A total of 407 valid responses were collected from young Chinese fast-food consumers with prior experience using AI-powered services through a structured questionnaire survey. Partial least squares structural equation modeling (PLS-SEM) was used to test the hypothesized relationships.

Findings – The empirical results reveal that perceived service quality ($\beta = 0.536, p < 0.001$) and perceived smart quality ($\beta = 0.148, p < 0.01$) both positively influence perceived experiential value ($R^2 = 0.419$), which subsequently enhances customer satisfaction ($\beta = 0.616, p < 0.001; R^2 = 0.380$) and habit formation ($\beta = 0.656, p < 0.001; R^2 = 0.430$). Satisfaction ($\beta = 0.369, p < 0.001$) and habit ($\beta = 0.413, p < 0.001; R^2 = 0.516$) both significantly promote customer loyalty. Among these relationships, the habit–loyalty path is strongest, highlighting habit’s dominant role in shaping loyalty.

Practical implications – Practitioners could combine efficient AI-enabled processes with human-centered service support to strengthen repeat patronage. Policymakers could improve data transparency, privacy protection and service standards to support responsible AI adoption.

Originality/value – This study develops an integrative framework that combines perceived smart quality with perceived service quality and explains loyalty formation in AI-enabled fast-food services through the parallel roles of satisfaction and habit.

Keywords Artificial intelligence, Fast-food restaurants, Perceived service quality, Perceived smart quality, Perceived experiential value, Customer loyalty

Paper type Research article

1. Introduction

In AI-driven fast-food restaurants, artificial intelligence (AI) is increasingly moving from back-end operations to frontline service encounters (Zemke *et al.*, 2020). AI-enabled kiosks, conversational agents, and service robots now participate in ordering, communication, service recovery, and service delivery (Ayalew *et al.*, 2025). As a result, customers often interact with both human staff and AI systems within a single dining episode, reflecting the growing prevalence of hybrid frontline service delivery (Kao and Huang, 2023). Although this hybrid model is becoming more common, it remains unclear how customer loyalty is formed when human-centered service cues and technology-mediated smart cues jointly shape customers’ experience evaluations and behavioral tendencies.

In classic service quality research, judgments of quality have often been based on the assumption that services are delivered by human employees (Parasuraman *et al.*, 1988). However, in AI-enabled frontline settings, research has increasingly examined hybrid human–AI service encounters, where human staff and AI systems jointly participate in service delivery (Zhang *et al.*, 2026). The adoption of AI technologies can enhance service efficiency and consistency, while also enabling more personalized and interactive customer experiences



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(Gordon *et al.*, 2026). Service robots not only perform repetitive tasks but also collaborate with employees to improve service performance, which reshapes frontline roles and customer interactions (Leño Calleja *et al.*, 2025). At the same time, evidence on heterogeneous customer responses, including generational differences, further suggests that hybrid service environments require clearer frameworks to explain how human and AI service cues jointly shape customer evaluations and loyalty-related behaviors (Wang *et al.*, 2024).

Although research on hybrid human–AI frontline services and the evolving roles of human employees and AI systems has expanded, most studies still emphasize either frontline automation or service augmentation (Raisch and Krakowski, 2021). As a result, the theoretical tension between human-centered and AI-mediated service quality remains underexplored. In addition, existing studies often examine human and AI service quality cues in isolation. They document the separate effects of these cues on customer satisfaction and loyalty, rather than modeling how they interact within the same service process. This fragmented approach overlooks the possibility that human-centered and AI-mediated quality cues may reinforce or offset each other when customers evaluate service encounters. Emerging evidence further shows that human–robot collaboration can shape service outcomes and influence customers' attribution and responsibility judgments (Zhang *et al.*, 2026). This issue is particularly salient in fast-food settings, where service encounters are frequent, highly standardized, and delivered within a short time span. Therefore, it is important to understand how human-centered and AI-mediated quality cues jointly influence customer satisfaction, habit formation, and loyalty.

To address this gap, we propose a dual quality perspective. We conceptualize perceived service quality as human-centered service quality, reflected in employee-delivered cues such as reliability, efficiency, and emotional support. We also introduce perceived smart quality to capture AI-mediated service attributes and interaction experiences enabled by intelligent systems (Azhar *et al.*, 2026; Wong *et al.*, 2022). Specifically, we conceptualize trend affinity, entertainment, and servicescape as three dimensions of perceived smart quality. Unlike hedonic quality, which primarily emphasizes sensory stimulation and immediate pleasure (Möhlmann, 2015), trend affinity in AI-enabled services reflects customers' psychological inclination to align with technological leadership and a modern digital lifestyle, rather than seeking sensory entertainment or esthetic enjoyment alone. Similarly, entertainment in AI-mediated services emphasizes interactive and engaging experiences that foster deeper customer participation through dynamic and responsive interfaces, going beyond passive enjoyment (Behera and Zhang, 2025). Finally, while traditional servicescape design focuses on the aesthetics and functionality of the physical environment (Bitner, 1992), servicescape in AI-enabled contexts extends to include both physical and digital elements shaped by intelligent technologies. Prior research suggests that AI-enabled servicescapes integrate digital interfaces into the service setting, creating a "smart servicescape" in which AI components reshape user interactions beyond the constraints of physical surroundings alone (Yin *et al.*, 2023; Wirtz *et al.*, 2018).

Driven by this dual-quality conceptualization, we argue that these service attributes do not operate in isolation but function as external stimuli that trigger internal psychological processing. Building on the human-centered logic of SERVQUAL and integrating the Stimulus–Organism–Response (S–O–R) paradigm with Habit Theory, we develop an integrative model. In this framework, human-centered service quality and perceived smart quality constitute the stimuli (S). The organism (O) is captured through customers' internal states, operationalized as experiential value, satisfaction, and habit formation, which ultimately shape customer loyalty as the response (R).

To empirically examine this theoretical mechanism within a real-world setting, China provides a relevant context because AI-enabled services are widely adopted and technology-intensive encounters are common, especially in the fast-food sector. This setting offers a robust opportunity to examine the proposed dual-quality mechanism in hybrid frontline services. Against this backdrop, this study pursues three specific objectives. First, we examine how human-centered service quality and perceived smart quality influence experiential value, and how experiential value subsequently shapes customer satisfaction and habit formation in

AI-driven fast-food restaurants. Second, we investigate how satisfaction and habit subsequently affect customer loyalty. Third, we test the mediating roles of experiential value, satisfaction, and habit in the relationships between the two quality constructs and customer loyalty.

2. Theoretical perspectives

2.1 Theoretical integration of SERVQUAL, SOR, and habit theory in AI-driven fast-food settings

Guided by the S–O–R paradigm, this study integrates SERVQUAL and habit theory to explain customer loyalty formation in AI-driven fast-food environments. SERVQUAL provides a foundational lens for conceptualizing human-centered service quality as customers' evaluations of employee-delivered service performance (Parasuraman *et al.*, 1988). Extending this logic to AI-enabled service settings, we conceptualize the stimulus (S) as two complementary quality cue systems: perceived service quality, derived from interpersonal service delivery, and perceived smart quality, derived from AI-mediated interactions and digitally supported service infrastructures.

To further develop the theoretical treatment of experiential value in AI-enabled services, we extend the S–O–R logic by deepening the organism (O) state. Although experiential value has traditionally been viewed as a holistic appraisal that integrates utilitarian performance and affective reactions, its formation may shift when service delivery becomes AI-mediated through automation and digital interfaces. In AI-enabled fast-food services, technology-mediated pathways reshape customers' psychological processing by redistributing perceived control and required effort. For example, AI-enabled self-service interfaces (e.g. kiosks and mobile ordering) can increase perceived agency by transferring action opportunities and process control to customers, thereby supporting more participatory and co-created service experiences (Puntoni *et al.*, 2021). This enhanced agency strengthens perceived autonomy and psychological empowerment, such that experiential value and subsequent satisfaction become more strongly rooted in perceived control rather than hedonic enjoyment alone (Larivière *et al.*, 2017). In parallel, automation can streamline service steps, reduce information-processing demands, and lower cognitive load. This allows customers to evaluate options with less mental effort and experience the encounter as smoother and more manageable (Grewal *et al.*, 2020). Together, these mechanisms clarify how experiential value functions as an enriched organism state in AI-enabled contexts and strengthen the coherence of our S–O–R-based model.

Finally, we draw on habit theory to explain the response (R) phase by linking customers' internal evaluations to routinized behavioral outcomes. In AI-mediated service contexts, satisfaction can translate into repeated patronage through the gradual development of automatic behavioral tendencies (Wood and Neal, 2009). Because intelligent systems typically provide standardized processes and low-friction interactions, repeated satisfactory experiences are reinforced more efficiently, which can accelerate the consolidation of stable consumption routines. Accordingly, customer loyalty in AI-enabled fast-food settings reflects not only conscious attitudinal commitment but also behavioral reinforcement through habit formation (Grewal *et al.*, 2020).

2.2 Hypothesis development

Drawing on these perspectives, this study proposes a model of six constructs. Figure 1 shows that loyalty in AI-driven fast-food settings arises from perceived service and perceived smart quality via experiential value, satisfaction, and habit.

2.2.1 Impact of perceived service quality and perceived smart quality on perceived experiential value. In service marketing, experiential value reflects customers' cognitive and affective interpretations of their service interactions (Karim and Mehzabeen, 2025). In AI-enabled fast-food services, customers are exposed to both interpersonal and technology-mediated value cues within the same service encounter. Therefore, we distinguish perceived

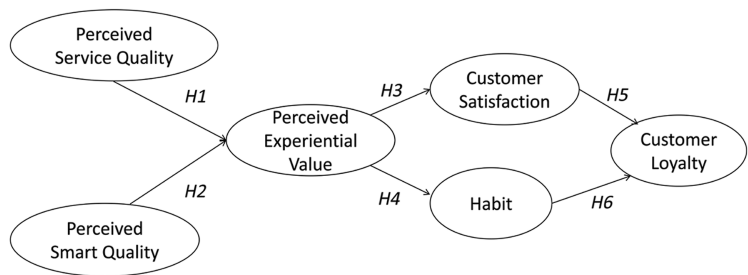


Figure 1. Conceptual model. Source: Developed by authors

service quality from perceived smart quality to avoid conceptual overlap between human-led and system-led performance. Perceived service quality contributes to experiential value by reducing service-related uncertainty and anxiety through employee professionalism, responsiveness, and emotional support (Min and Min, 2011). In contrast, perceived smart quality enhances experiential value through technology-enabled attributes, including trend affinity, entertainment, and an AI-enabled servicescape that integrates digital interfaces into the service setting (Asghar et al., 2024). Rather than treating human and smart service cues as substitutes, we argue that they operate as complementary stimuli. Perceived service quality provides a reliable service foundation, whereas perceived smart quality delivers modern, efficient, and interactive elements that increasingly shape customer value perceptions in AI-enabled fast-food environments.

We present the following hypotheses.

- H1a. Perceived service quality influences perceived experiential value positively.
- H1b. Perceived smart quality influences perceived experiential value positively.

2.2.2 *The impact of perceived experiential value on satisfaction and habit.* Within the S–O–R framework, experiential value represents the internal organism (O) state that connects service perceptions to downstream behavioral outcomes. Prior research often emphasizes satisfaction as the primary outcome of positive service experiences (Cheng et al., 2020). However, focusing solely on satisfaction may provide an incomplete understanding of customer behavior, particularly in fast-food settings where decisions are made quickly and repeated frequently. We argue that experiential value can activate two psychological pathways: an evaluative route reflected in satisfaction and an automatic route reflected in habit formation. Satisfaction involves a conscious appraisal of whether the service experience meets expectations. In contrast, habit refers to a learned automatic tendency that develops when service encounters are consistently smooth and require minimal effort (Wood and Neal, 2007). Although habit formation is traditionally viewed as a gradual process, AI-mediated service delivery may accelerate it by reducing cognitive effort and streamlining decision steps. For example, AI-enabled tools that simplify ordering and payment can lower information-processing demands, allowing a positive experience to transition into a stable routine more quickly (Yoo and Cho, 2020).

The following hypotheses are proposed.

- H2a. Perceived experiential value positively influences customer satisfaction in AI-driven fast-food environments.
- H2b. Perceived experiential value positively impacts customer habits in AI-driven fast-food environments.
- H3a. Customer satisfaction positively influences customer loyalty in AI-driven fast-food environments.

H3b. Customer habit positively influences customer loyalty in AI-driven fast-food environments.

2.2.3 The mediating effect of experiential value, satisfaction, and habit. To understand how service quality translates into long-term loyalty, it is necessary to examine the psychological mechanisms through which these effects unfold. We therefore propose a mediation framework in which experiential value serves as an immediate appraisal that activates downstream responses reflected in satisfaction and habit-related tendencies. This structure is particularly relevant in AI-enabled fast-food services, where customers may seek interpersonal warmth while simultaneously valuing technology-enabled efficiency, and these dual cues can shape loyalty through distinct mechanisms (Khan *et al.*, 2022). Specifically, we model sequential mediation, assuming that experiential value precedes satisfaction and habit formation because value perceptions provide the foundation for subsequent psychological evaluation and behavioral continuity. This framework clarifies how service encounters can generate both evaluative and routinized pathways that jointly support more stable customer loyalty (Lemon and Verhoef, 2016).

H4a. Perceived experiential value positively mediates the pathway between perceived service quality and customer satisfaction.

H4b. Perceived experiential value positively mediates the pathway between perceived service quality and customer habit.

H4c. Perceived experiential value positively mediates the pathway between perceived smart quality and customer satisfaction.

H4d. Perceived experiential value positively mediates the pathway between perceived smart quality and customer habit.

H4e. Perceived experiential value and customer satisfaction sequentially and positively mediate the pathway between perceived service quality and customer loyalty.

H4f. Perceived experiential value and habit sequentially and positively mediate the pathway between perceived service quality and customer loyalty.

H4g. Perceived experiential value and customer satisfaction sequentially and positively mediate the pathway between perceived smart quality and customer loyalty.

H4h. Perceived experiential value and habit sequentially and positively mediate the pathway between perceived smart quality and customer loyalty.

H4i. Customer satisfaction positively mediates the pathway between perceived experiential value and customer loyalty.

H4j. Customer habit positively mediates the pathway between perceived experiential value and customer loyalty.

In summary, drawing on SERVQUAL, the SOR paradigm, and habit theory, we conceptualize perceived service quality and perceived smart quality as dual stimuli that shape satisfaction and habit-related tendencies, ultimately influencing customer loyalty. The following section details the research design, measurement approach, and PLS-SEM procedures used to test these hypotheses empirically.

3. Method

3.1 Ethical approval

This study complied with institutional and national ethical standards. According to the authors' institutional policy, survey-based research that does not involve sensitive information or vulnerable groups is exempt from ethics review. All procedures adhered to relevant

guidelines and regulations. Informed consent was obtained from all participants: (1) the purpose of the study was clearly explained; (2) anonymity and confidentiality were ensured, with no identifiable information collected and responses analyzed in aggregate; (3) data were securely stored and used solely for academic research; and (4) participants were informed that the study involved no foreseeable risks and that they could withdraw at any time without consequence.

3.2 Sample and data collection

Data were collected via WJX.cn, a major Chinese online survey platform. Using purposive sampling, we targeted consumers aged 18 to 35 who had prior experience with AI-enabled services. This sampling strategy was adopted for two reasons. First, the study focuses on AI-enabled fast-food service encounters, and respondents needed direct exposure to AI features such as self-service ordering, automated service processes, or intelligent interfaces to provide valid evaluations. Second, younger consumers are generally more familiar with digital technologies and more willing to adopt emerging service modes, making them more likely to engage with AI-enabled restaurant services in practice (Verhoef *et al.*, 2021). Therefore, purposive sampling through WJX was considered appropriate for reaching the most relevant user group aligned with the study objectives. However, given the online recruitment context, the findings should be interpreted as reflecting the perceptions and behavioral tendencies of young, digitally engaged fast-food customers, and caution is warranted when generalizing the results to broader demographic groups. Before starting the questionnaire, respondents were provided with a standardized description of AI-driven restaurant features (e.g. AI ordering systems, smart tables, and robotic delivery) and were shown representative photographs to ensure a consistent understanding of the scenario. After removing invalid responses, 407 valid questionnaires were retained, exceeding the minimum sample size requirement for PLS-SEM and ensuring adequate statistical power (Hair *et al.*, 2022). Respondent characteristics are reported in Table 1.

3.3 Measures

All measures were adapted from validated scales (see Table A1 in the Appendix) and covered six constructs: perceived service quality, perceived smart quality, experiential value,

Table 1. Profile of respondents (N = 407)

Characteristics		Frequency	Percentage
Gender	Male	185	45.5
	Female	184	45.2
	Others	38	9.3
Age	18–20	130	31.9
	21–23	159	39.1
	24–26	86	21.1
	27–35	32	7.9
Education	Up to high school level	2	0.5
	Two-year college degree	89	21.9
	Undergraduate degree	229	56.3
	Graduate degree	64	15.7
	Doctoral degree	23	5.7
Monthly income	¥ 3,000 or below	58	14.3
	¥3,000 - ¥5,999	119	29.2
	¥6,000 - ¥10,000	186	45.7
	¥10,000 or above	44	10.8
Source(s): Developed by authors			

satisfaction, habit, and loyalty. Perceived service quality and perceived smart quality were modeled as second-order reflective constructs, with their first-order dimensions specified as reflective constructs and their measurement items modeled as reflective indicators. Under this reflective specification, the higher-order construct is manifested through its first-order dimensions, such that the overall evaluation is consistently reflected across related dimensions (Jarvis *et al.*, 2003). This modeling choice is appropriate because customers typically form a holistic impression of service encounter quality and AI-mediated service experiences, and their evaluations tend to be coherent across dimensions. Accordingly, changes in the higher-order construct are expected to be systematically reflected in its lower-order dimensions (Hair *et al.*, 2022).

All measurement items were translated into Chinese using a translation–back translation procedure and reviewed by three scholars to ensure semantic equivalence and cultural appropriateness. A small pretest ($n = 30$) was then conducted to confirm clarity and comprehension, after which minor wording refinements were made before administering the final survey. The final instrument included 40 items. A 7-point Likert scale (1 = strongly disagree, 7 = strongly agree) was used. Demographics captured gender, age, education, and familiarity with AI.

3.4 Data analysis

This study adopted a quantitative approach and applied partial least squares structural equation modeling (PLS-SEM) (Bui and Tran, 2024) to conduct the empirical analysis and test the proposed research model and hypotheses. Data were analyzed in SmartPLS v.4, a widely adopted software for PLS-SEM, to assess the measurement and structural models. PLS-SEM was selected due to its ability to handle complex models with multiple constructs and mediating relationships. First, the proposed framework involves multiple interrelated constructs and mediation mechanisms that need to be estimated simultaneously, and PLS-SEM supports the testing of both direct and indirect effects through bootstrapping. Second, PLS-SEM provides flexibility for modeling multidimensional constructs using a higher-order structure, allowing the joint assessment of measurement quality and structural relationships within a single analytical framework. Third, given the study's focus on an emerging AI-enabled service setting, PLS-SEM is well suited for evaluating the model's explanatory and predictive performance for key outcomes (Shmueli *et al.*, 2019).

4. Results

4.1 Measurement model

Before hypothesis testing, we conducted diagnostic checks to verify the dataset's suitability for PLS-SEM, covering common method bias, multicollinearity, and distributional properties. Common method bias was assessed using the full collinearity approach (Kock, 2015), in which each latent construct was regressed on a randomly generated marker variable and the resulting full-collinearity VIFs were inspected with all values below 3.3 (see Table A2 in Appendix). Multicollinearity in the structural model was examined using standard inner VIFs in SmartPLS 4.0, which ranged from 1.426 to 3.564 and were below the 5.0 cutoff (O'Brien, 2007). For normality, skewness ranged from -1.716 to 0.058 and most kurtosis values fell between -1.06 and 1.30 , with a few slightly above 3, indicating mild leptokurtosis. These deviations are acceptable because PLS-SEM does not require normally distributed data (Hair *et al.*, 2022).

For model estimation, we applied the disjoint two-stage approach in SmartPLS 4. Measurement model evaluation covered reliability, convergent and discriminant validity, and multicollinearity. As reported in Table 2, all constructs met recommended thresholds: Cronbach's alpha and composite reliability (CR) were ≥ 0.70 (Hair *et al.*, 2022), average variance extracted (AVE) exceeded 0.50 (Fornell and Larcker, 1981), all outer loadings were

Table 2. Measurement model results

Construct	Items	Mean scores	Factor loading	Cronbach's Alpha	CR	AVE
Consistency	CO1–CO4	5.055	0.877–0.890	0.906	0.934	0.780
Empathy	EM1–EM4	5.386	0.872–0.892	0.906	0.934	0.780
Efficiency	EF2–EF4	5.497	0.917–0.924	0.912	0.944	0.850
Assurance	AS2–AS4	4.947	0.885–0.920	0.887	0.93	0.815
Trend Affinity	TA2–TA4	5.183	0.888–0.928	0.891	0.932	0.821
Entertainment	EN2–EN4	5.509	0.909–0.915	0.9	0.937	0.833
Servicescape	SS2–SS4	4.835	0.894–0.904	0.881	0.927	0.808
Perceived Experiential Value	PEV1–PEV4	5.051	0.757–0.821	0.816	0.874	0.635
Customer Satisfaction	CS1–CS3	5.366	0.849–0.889	0.837	0.902	0.754
Habit	HA1–HA3	5.440	0.837–0.897	0.849	0.909	0.769
Customer Loyalty	CL1–CL3	4.985	0.879–0.897	0.863	0.916	0.785
Perceived Service Quality	–	–	–	0.930	0.939	0.525
Perceived Smart Quality	–	–	–	0.891	0.912	0.534

Source(s): Developed by authors

>0.70. Discriminant validity was supported, with HTMT values below the conventional cutoff of 0.90 and the Fornell-Larcker criterion satisfied (Table A3 in the Appendix; Ab Hamid *et al.*, 2017). This indicates that constructs such as human-centered service quality and perceived smart quality are empirically distinct, with limited evidence of conceptual overlap. Table A4 further shows that the second-order reflective constructs (PSQ and PSQsmart) satisfied measurement requirements: VIFs for all first-order dimensions were <3.1, outer loadings ranged from 0.793 to 0.826, and all were significant ($p < 0.001$). Overall, these results confirm adequate discriminant validity, supporting interpretation of the structural paths as relationships among distinct constructs rather than measurement overlap.

4.2 Test of research hypotheses

To test the hypothesized relationships, the structural relationships were tested using a bootstrapping resampling approach implemented in SmartPLS (5,000 subsamples). In this model, perceived service quality and perceived smart quality were used to capture the hierarchical structure of customer perceptions. The measurement quality of these higher-order constructs had been confirmed in the measurement model.

Path coefficients and their corresponding t - and p -values were analyzed to test the hypotheses. Figure 2 and Table A5 show that perceived service quality significantly influenced perceived experiential value ($\beta = 0.536$, $t = 10.608$, $p < 0.001$), supporting H1a. This effect was more than three times larger than that of perceived smart quality, suggesting that customers' positive experiences are still primarily shaped by reliable and empathetic human service. A plausible reason is that fast-food encounters are frequent and time-pressured, so customers value quick responses and interpersonal reassurance when forming value perceptions. Similarly, perceived smart quality had a significant effect on perceived experiential value ($\beta = 0.148$, $t = 2.813$, $p < 0.01$), supporting H1b. However, its comparatively smaller coefficient suggests a supplementary role, implying that AI-mediated functionalities currently add incremental value mainly through convenience and efficiency, building on the baseline experience established by frontline employees. Taken together, these findings suggest a differentiated pattern of value formation, in which human-centered service quality acts as the dominant driver, whereas smart quality contributes as a supportive factor.

In turn, perceived experiential value significantly affected customer satisfaction ($\beta = 0.616$, $t = 18.044$, $p < 0.001$) and habit ($\beta = 0.656$, $t = 20.301$, $p < 0.001$), supporting H2a and H2b. Notably, experiential value showed a slightly stronger effect on habit

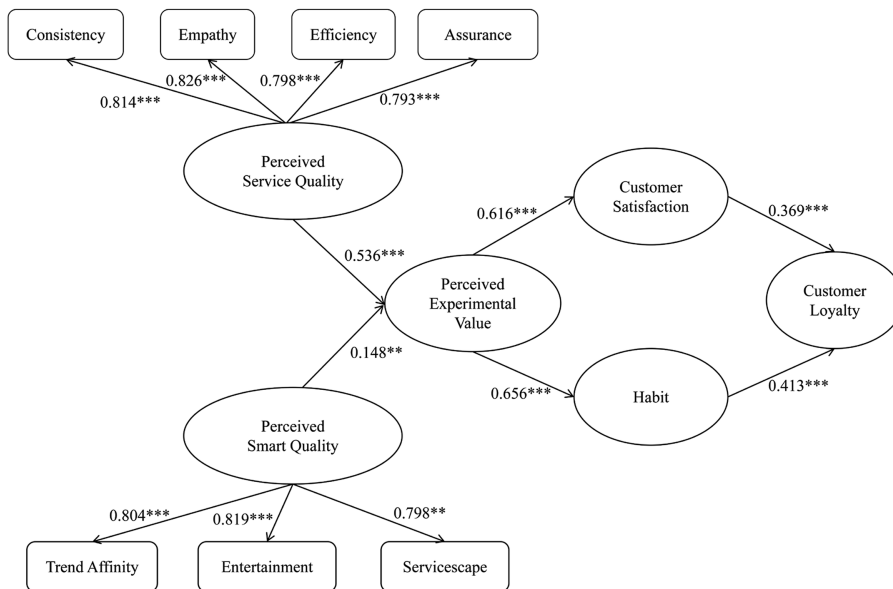


Figure 2. Structural model results: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. Source: Developed by authors

than on satisfaction, indicating that value-rich experiences in AI-enabled fast-food services are not only evaluated positively but are also more likely to be internalized into repeated behavioral patterns. From a behavioral economics perspective, this robust value-to-habit linkage is consistent with repeated-choice dynamics and reinforcement learning. When customers repeatedly experience immediate functional and affective benefits, such as convenience, efficiency, and a smooth service flow shaped by human-centered support and AI-mediated interactions, these benefits strengthen the tendency to repeat the same choice and reduce reliance on deliberate decision-making, thereby promoting behavioral automaticity.

Finally, both customer satisfaction ($\beta = 0.369$, $t = 6.085$, $p < 0.001$) and habit ($\beta = 0.413$, $t = 6.653$, $p < 0.001$) positively influenced customer loyalty, supporting H3a and H3b. Notably, habit formation exerted a stronger effect on loyalty than satisfaction, suggesting that loyalty formation in AI-enabled fast-food settings may depend more on routinized repetition than on episodic evaluative judgments. Satisfaction reflects customers' reflective assessments of service encounters, which can vary across occasions, whereas habit formation captures more stable, cue-triggered behavioral tendencies reinforced through repeated use. In high-frequency and low-involvement contexts, AI-mediated features such as standardized ordering procedures and reduced effort can further stabilize routines, making loyalty more persistent even when satisfaction fluctuates. Taken together, these findings indicate that experiential value strengthens both satisfaction and habit formation, while habit formation represents a particularly stable pathway linking service experiences to long-term loyalty.

To assess the model's ability to account for the variance in the dependent constructs, R^2 values were examined. The explained variance for each endogenous variable was as follows: perceived experiential value ($R^2 = 0.419$), customer satisfaction ($R^2 = 0.380$), habit ($R^2 = 0.430$), and customer loyalty ($R^2 = 0.516$). According to Chin (1998), an R^2 value exceeding 0.33 suggests a moderate level of explanatory power, whereas values above 0.67 denote a strong or substantial capacity to explain variance.

Effect sizes (f^2) were further analyzed to compare the relative contribution of each predictor (Cohen, 2013). Perceived service quality had a medium-to-large effect on experiential value

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($f^2 = 0.25$), whereas perceived smart quality had only a small effect ($f^2 = 0.02$), confirming that reliable and empathetic human service remains more influential than technological attributes in shaping customer experience. Experiential value exerted large effects on both satisfaction ($f^2 = 0.61$) and habit ($f^2 = 0.75$), showing its central role in driving customers' emotional and behavioral responses within AI-driven restaurants. Finally, satisfaction ($f^2 = 0.15$) and habit ($f^2 = 0.19$) had moderate effects on loyalty, with habit showing a slightly stronger impact. Taken together, these findings indicate that while AI-mediated features can enhance the service encounter, human-centered service quality and the resulting habit formation represent the primary pathways sustaining customer loyalty.

The predictive performance of the PLS-SEM model was assessed using the PLS prediction approach (Shmueli et al., 2019). $Q^2_{predict}$ values above zero indicate predictive relevance, and differences in mean absolute error (MAE) and root mean square error (RMSE) were computed as PLS-SEM values minus the indicator average (IA) benchmark, with negative values showing superior performance. As shown in Figure A1, all $Q^2_{predict}$ values (0.117–0.708) exceeded zero, and except for one measurement, RMSE and MAE were lower than the IA benchmark, confirming strong predictive accuracy. Thus, the model not only fits the theoretical framework but also demonstrates robust predictive power for practical applications.

This study used SmartPLS 4 to conduct bootstrapping with 5,000 resamples to test indirect effects. As shown in Table 3, all ten mediation hypotheses (H4a–H4j) were supported. Perceived experiential value significantly mediated the relationships between both perceived service quality and perceived smart quality and customer satisfaction (H4a, H4c) and customer habit (H4b, H4d). However, the indirect effects of service quality ($\beta = 0.330$ – 0.351) were notably stronger than those of perceived smart quality ($\beta = 0.091$ – 0.097), suggesting that reliable, empathetic human service generates deeper experiential value and more sustained behavioral engagement than technology-based features alone. Sequential mediation via perceived experiential value and customer satisfaction (H4e, H4g) or customer habit (H4f, H4h) was also confirmed. Finally, perceived experiential value indirectly influenced customer loyalty through satisfaction (H4i) and habit (H4j), highlighting its pivotal role in translating both human and AI service quality into long-term loyalty. These mediation results further clarify how customer experiences in AI-driven fast-food restaurants evolve from initial service perceptions to emotional attachment and finally to enduring loyalty behaviors.

In summary, the structural paths and mediation tests jointly indicate that dual quality cues influence customer loyalty through satisfaction and habit-related mechanisms, clarifying how service experiences translate into enduring behavioral outcomes. Building on these findings, the next section discusses the theoretical interpretations and practical implications of AI-enabled fast-food services.

Table 3. Results of mediating tests

Hypothesis	Paths	Path coefficients	<i>t</i> -value	<i>p</i> -value	Results
H4a	PSQ → PEV → CS	0.330	8.268	0.000	Supported
H4b	PSQ → PEV → HA	0.351	8.244	0.000	Supported
H4c	PSQsmart → PEV → CS	0.091	2.750	0.006	Supported
H4d	PSQsmart → PEV → HA	0.097	2.795	0.005	Supported
H4e	PSQ → PEV → CS → CL	0.122	4.523	0.000	Supported
H4f	PSQ → PEV → HA → CL	0.145	4.586	0.000	Supported
H4g	PSQsmart → PEV → CS → CL	0.034	2.368	0.018	Supported
H4h	PSQsmart → PEV → HA → CL	0.040	2.486	0.013	Supported
H4i	PEV → CS → CL	0.227	5.146	0.000	Supported
H4j	PEV → HA → CL	0.271	5.651	0.000	Supported

Source(s): Developed by authors

5. Discussion and conclusions

Our findings show that human-centered service quality exerts a stronger influence on customer outcomes than perceived smart quality. This pattern is consistent with evidence from Asian markets, where empathy and responsiveness remain salient evaluation cues even in technology-rich service environments (Chen *et al.*, 2025; Hwang *et al.*, 2022). However, it differs from findings in highly automated or efficiency-oriented settings, such as robotic restaurants, where automation competence, novelty, and interactive features play a more prominent role in shaping customer responses (Ali *et al.*, 2025; Otemisova *et al.*, 2025). This divergence likely reflects contextual differences. In hybrid human–AI fast-food services, customers often engage in frequent, low-involvement encounters and may rely more on cues that reduce uncertainty, making interpersonal reassurance particularly diagnostic. Moreover, the perceived novelty of AI-mediated features may diminish with repeated use, especially among younger consumers, as smart functionalities become a baseline expectation rather than a differentiator. Taken together, the relative importance of human-centered versus smart quality cues appears to be context dependent, shaped by the degree of service hybridity and customers' evolving expectations.

5.1 Conclusions

This study offers three key insights. First, perceived service quality remains the dominant driver of customer outcomes in AI-enabled fast-food services, underscoring the continued importance of human-centered service cues. Second, loyalty is supported through two complementary routes: experiential value strengthens both satisfaction and habit formation, with habit formation showing a stronger link to loyalty. Third, AI does not replace human service; when integrated into human-centered design, AI-mediated features can enhance convenience and interaction smoothness, helping brands sustain long-term customer relationships during digital transformation.

5.2 Theoretical implications

First, this study contributes to the AI service quality literature by proposing an integrative framework that combines perceived smart quality with perceived service quality in AI-enabled fast-food services. Although prior research has increasingly examined AI applications in frontline service encounters, much of it remains focused on customers' initial attitudes toward technology use or broad behavioral reactions to automation (Gonçalves *et al.*, 2024; Koo *et al.*, 2023). By conceptualizing service quality as a dual-source evaluation system, this study clarifies how AI-mediated smart attributes and human-centered service cues jointly shape customer loyalty, rather than operating as independent predictors. Moreover, by identifying satisfaction and habit formation as concurrent mediating mechanisms, our framework offers a more fine-grained explanation of how quality perceptions translate into long-term loyalty in AI-enabled service contexts.

Second, this study extends the S–O–R paradigm by developing a framework tailored to AI-enabled frontline service encounters. Rather than treating service quality as a single stimulus, we conceptualize the stimulus stage as a dual-source structure that distinguishes between human-centered service cues (captured through SERVQUAL dimensions) and AI-mediated smart cues (captured through perceived smart quality). At the organism stage, we specify customers' internal processing as coexisting psychological states, operationalized through experiential value, satisfaction, and habit formation, which translate hybrid stimuli into downstream outcomes. The response stage is operationalized as customer loyalty, shifting the analytical focus from immediate transactional intentions to more enduring behavioral outcomes. Overall, this framework clarifies how mixed human–technology service cues are internalized in AI-enabled service settings and broadens the applicability of the S–O–R model in emerging frontline contexts (Akram *et al.*, 2020; Yang *et al.*, 2021).

Third, this study contributes to habit theory by demonstrating its critical role in driving customer loyalty in AI-enabled services. By examining hybrid frontline service delivery, the findings show how human-centered and AI-mediated service cues jointly shape loyalty formation. A key empirical result is that habit formation exerts a stronger influence on customer loyalty than satisfaction in AI-enabled fast-food settings, indicating a shift toward more automatic behavioral routes (Chinelato and Gonçalves Filho, 2026). Specifically, human interaction can provide reassurance and reduce uncertainty, whereas standardized and low-friction AI-mediated processes reduce the effort required for decision-making. Over repeated encounters, these conditions can weaken customers' reliance on continuous evaluative judgments and strengthen routinized loyalty grounded in behavioral reinforcement.

5.3 Practical implications

Based on the empirical findings, this study offers specific recommendations for both industry practitioners and policymakers to optimize AI integration in the fast-food sector.

Fast-food practitioners should deploy adaptive AI-enabled interfaces to encourage repeat use and strengthen customers' default choice tendencies (Suttikun and Mahasuweerachai, 2025). Our findings indicate that habit-related mechanisms represent an important pathway for sustaining long-term loyalty. Rather than relying on static digital menus, restaurants can embed adaptive prompts that respond to customers' prior behaviors. For example, systems can recognize returning customers, prioritize frequently ordered items, and enable one-click reordering and checkout. During peak hours, a streamlined "fast-order" pathway can further reduce perceived effort and waiting-related frustration. By lowering cognitive load and interaction friction, these AI-mediated design features can make ordering feel easier and more familiar over time, thereby supporting more stable loyalty outcomes.

Managers should also establish a human-AI collaborative service model to ensure that efficiency gains do not come at the expense of the emotional quality of service encounters. AI can effectively handle standardized and high-frequency tasks such as ordering, payment, and routine inquiries. However, customers still value human-centered elements such as warmth, empathy, and personalized recognition, which remain difficult to fully replicate through automated systems. Practically, restaurants can reallocate frontline staff time toward high-empathy touchpoints, including proactive hospitality, exception handling, and complex complaint resolution. This division of labor enables firms to benefit from AI-enabled efficiency while preserving relational value, thereby strengthening customer loyalty in AI-enabled fast-food settings.

Policymakers can facilitate responsible AI adoption in the fast-food industry by strengthening governance frameworks. First, they should enhance data privacy and transparency in AI-enabled personalization by clarifying consent procedures, standardizing disclosures of data-use purposes, and providing convenient opt-out mechanisms. These measures can reduce customers' perceived data-related risks and reinforce trust in AI-enabled services. Second, policymakers should promote industry capacity building and service standardization by supporting frontline training for human-AI collaboration and establishing baseline requirements for AI interfaces, including usability, accessibility, reliability, and fairness. In doing so, the industry can improve operational efficiency while safeguarding customer rights and maintaining service quality.

5.4 Limitations and future research

The findings advance understanding of how perceived service quality and perceived smart quality shape satisfaction, habit formation, and loyalty in AI-enabled fast-food services. Several limitations suggest directions for future research. First, the sample was recruited purposively via WJX and consisted largely of younger consumers from metropolitan areas, which may limit representativeness. Future studies should include broader regions and age groups and adopt probability-based or mixed recruitment strategies to enhance external

validity. Second, given the fast-food focus, future research could extend the model to full-service and fine-dining contexts and conduct cross-country comparisons to examine cultural influences and variation in AI diffusion. Third, the cross-sectional design supports associative rather than causal interpretation and cannot capture the temporal dynamics of habit formation; longitudinal designs or contextualized experiments could provide stronger causal evidence and clarify cohort differences. Finally, future studies should further address responsible AI issues in AI-mediated services, including algorithmic transparency, privacy governance, and perceived bias, as well as potential risks such as reduced social interaction and unequal access for older or less digitally skilled users. Future work could also link these mechanisms to business outcomes such as customer lifetime value and cost-to-serve reduction.

Appendix

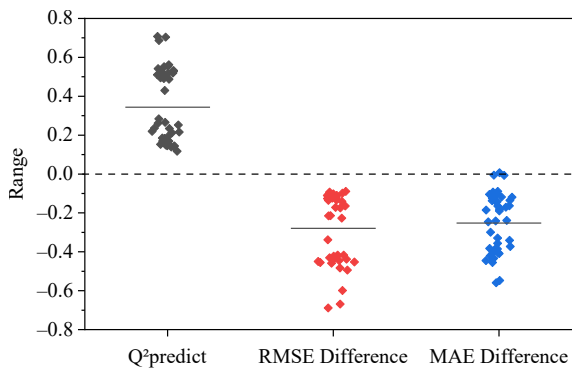


Figure A1. Results of Q^2 predict, RMSE, and MAE differences (PLS-SEM – IA). Source: Developed by authors

Table A1. List of measuring constructs

Constructs	Theoretical definitions	Operational definition	Items	Scope conditions	Sources
Trend affinity (TA)	The tendency to follow trends and adopt innovative services. (Source: Möhlmann, 2015)	Visiting a smart restaurant enhances trendiness and reflects alignment with current dining trends	TA1: This smart restaurant is a new trend I want to try TA2: People will consider me fashionable if I dine at this smart venue TA3: Visiting this smart restaurant will make me feel up-to-date TA4: Dining at this smart restaurant shows that I follow the latest dining trends	Applies to AI-driven fast-food settings where choices reflect technological novelty and trend awareness	Möhlmann (2015) So <i>et al.</i> (2018)

(continued)

Table A1. Continued

Constructs	Theoretical definitions	Operational definition	Items	Scope conditions	Sources
Entertainment (EN)	The leisure experiences that provide pleasure, relaxation, and revitalization. (Source: Kouzmál, 2020)	Using AI-driven entertainment at a smart restaurant enhances entertainment, engagement, and interest in the restaurant experience	EN1: I had fun using the AI system at this smart restaurant EN2: I enjoyed participating in the entertainment activities such as the lottery generated by the AI system at this smart restaurant EN3: I gained significant enjoyment from the AI experience at this smart restaurant EN4: The form of AI entertainment stimulated my interest in this restaurant experience	Applies to AI-driven fast-food settings where AI systems create interactive and enjoyable experiences	Kouzmál (2020) Koo et al. (2023)
Servicescape (SS)	The physical setting of service delivery, encompasses elements like color, music, scent, and spatial design. (Source: Bitner, 1992)	The atmosphere and features of AI-driven systems in a smart restaurant, such as smart ordering devices, digital menu boards, and AI assistants, create a cheerful, impressive, and personalized experience	SS1: The atmosphere of using AI systems in the restaurant such as smart ordering devices was cheerful SS2: The decor of this smart restaurant such as the digital menu board was fashionable SS3: The interface of the AI assistant was neat and friendly SS4: This AI system of the restaurant gave customers personal attention	Applies to smart restaurant environments featuring AI-enabled physical and digital design elements	Park et al. (2019) Yin et al. (2023)
Consistency (CO)	The consistent and uniform delivery of services regardless of time, place, or service provider. (Source: Herath et al., 2019)	A smart restaurant realizes consistent standards of food quality, maintains standardized processes, ensures consistent service quality, and guarantees stability in order processing and delivery	CO1: Smart restaurants would realize stable and reliable food quality CO2: The processes in smart restaurants is standardized CO3: Smart restaurant service systems ensure consistent service quality CO4: Smart restaurant systems ensure consistency in order processing and delivery	Applies to AI-driven service processes emphasizing stable quality and standardized operations	Herath et al. (2019)

(continued)

Table A1. Continued

Constructs	Theoretical definitions	Operational definition	Items	Scope conditions	Sources
Empathy (EM)	The ability to see things from others' perspectives. (Source: Lv et al. (2022))	This smart restaurant considers customer preferences, responds to their needs, provides individualized attention, and utilizes AI systems designed with customers' best interests at heart	EM1: I think this smart restaurant takes my preferences EM2: I think that this smart restaurant is receptive to my needs EM3: This smart restaurant gives its customers individualized attention EM4: The AI systems in the restaurant have the customers' best interests at heart	Applies to AI-assisted services designed to recognize and respond to individual customer needs	Lv et al. (2022) Ali et al. (2021)
Efficiency (EF)	The ability to respond to customer needs while improving service performance. (Source: Raub and Liao, 2012)	The system in the smart restaurant facilitates finding desired dishes, navigating different sections like the menu and order history, completing orders efficiently, and receiving food quickly	EF1: The system in the smart restaurant makes it easy for me to find the dishes I need EF2: It is easy to navigate through different sections in the smart restaurant EF3: The system enables me to complete an order quickly EF4: Smart restaurant would allow me to receive food promptly	Applies to AI-based service systems enhancing speed, accuracy, and convenience in fast-food operations	Raub and Liao (2012) Huang et al. (2023)
Assurance (AS)	Trust in the provider's expertise and the safety and quality of the service provided. (Source: Parasuraman et al., 1988)	Ensures timely service, builds customer trust, and responds efficiently to customer needs	AS1: This smart restaurant provides meals on schedule AS2: This smart restaurant inspires customer trust AS3: This smart restaurant is consistently responsive AS4: This smart restaurant is capable of addressing a customer's query promptly	Applies to AI-driven fast-food contexts where customers rely on system reliability and staff expertise for secure and trustworthy service	Gabrow (2021)

(continued)

Table A1. Continued

Constructs	Theoretical definitions	Operational definition	Items	Scope conditions	Sources
Perceived Experiential Value (PEV)	The benefit that comes from a person using and experiencing a good or service. (Source: Turel et al., 2010)	The worth gained from engaging with AI services in a smart restaurant, where effort, price, and interaction enhance enjoyment and usefulness	PEV1: For the involvement and effort expended on the restaurant, I feel it is valuable PEV2: For the price and effort expended by visiting the smart restaurant, I feel it is worthwhile PEV3: My involvement in participating AI system in the smart restaurant makes the service interactions more enjoyable PEV4: Using a smart restaurant App enhances my effectiveness in managing my personal life	Applies to AI-driven fast-food settings where enjoyment and usefulness arise from interacting with intelligent systems	Kim and Tang (2020) Turel et al. (2010)
Habit (HA)	The repeated and consistent practices formed through routine engagement. (Source: Shove, 2012)	Frequent use of AI ordering, smart tables, or robot delivery has become a routine part of dining	HA1: Using the AI ordering system/Smart table/Robot delivery at the smart restaurant has become part of my routine HA2: I must use the AI ordering system/Smart table/Robot delivery when I visit the smart restaurant HA3: Using the AI ordering system/Smart table/Robot delivery has become natural to me	Applies to repetitive dining behaviors formed through routine use of AI ordering and service technologies	Venkatesh et al. (2003)
Customer Satisfaction (CS)	Emotional response to meeting or missing expectations. (Source: Kotler and Keller, 2012)	Fulfillment with its products and services, successful persuasion, and the achievement of expectations	CS1: I am pleased with the offerings and service quality of this smart restaurant CS2: This smart restaurant successfully convinced me CS3: This smart restaurant has met my expectations	Applies to AI-driven fast-food environments where customer expectations are shaped by human-AI service integration	Saneva and Chortoseva (2022)

(continued)

Table A1. Continued

Constructs	Theoretical definitions	Operational definition	Items	Scope conditions	Sources
Customer Loyalty (CL)	The tendency to repurchase or continue using a brand or service. (Oliver, 1999)	Willingness to recommend and revisit the smart restaurant frequently	CL1: I would suggest this smart restaurant to others CL2: I would visit this smart restaurant frequently CL3: If I have to go fast food restaurant again, I would be sure to visit this store again	Applies to AI-enhanced fast-food settings where continued patronage reflects satisfaction and habitual engagement with smart services	Saneva and Chortoseva (2022)

Note(s): EF1, AS1, TA1, EN1, and SS1 were removed based on the measurement model
Source(s): Developed by authors

Table A2. Common method bias assessment via full collinearity test

Construct	AS	CL	CO	CS	EF	EM	EN	HA	PEV	SS	TA
VIF	2.682	2.211	2.713	2.022	3.095	2.680	2.797	2.176	1.865	2.460	2.799

Source(s): Developed by authors

Table A3. Results of the HTMT analysis for discriminant validity evaluation

	CO	EM	EF	AS	TA	EN	SS	PEV	CS	HA	CL
CO											
EM	0.664										
EF	0.563	0.591									
AS	0.605	0.588	0.613								
TA	0.567	0.619	0.543	0.610							
EN	0.464	0.572	0.568	0.488	0.526						
SS	0.343	0.48	0.396	0.307	0.538	0.540					
PEV	0.517	0.532	0.617	0.613	0.430	0.603	0.305				
CS	0.507	0.591	0.583	0.612	0.517	0.593	0.385	0.697			
HA	0.553	0.601	0.579	0.597	0.598	0.609	0.351	0.736	0.810		
CL	0.539	0.647	0.647	0.602	0.602	0.725	0.433	0.749	0.765	0.779	

Note(s): As perceived service quality and perceived smart quality are operationalized as second-order constructs via the repeated indicators method, HTMT ratios are not applicable for assessing their discriminant validity. Accordingly, these values are omitted from the results

Source(s): Developed by authors

Table A4. Assessment of second-order reflective constructs

Second-order constructs	First-order constructs	VIF	Outer loading	p-value
Perceived Service Quality (PSQ)	Consistency	2.784	0.814	<0.001
	Empathy	2.546	0.826	<0.001
	Efficiency	3.097	0.798	<0.001
	Assurance	2.764	0.793	<0.001
Perceived Smart Quality (PSQsmart)	Trend Affinity	2.503	0.804	<0.001
	Entertainment	2.831	0.819	<0.001
	Servicescape	2.898	0.798	<0.001
Source(s): Developed by authors				

Table A5. Summary of key path coefficients, effect sizes, and hypothesis-testing results

Hypothesis	Paths	Path coefficients	t-value	p-value	f ²	Results
H1a	PSQ → PEV	0.536	10.608	0.000	0.253	Supported
H1b	PSQsmart → PEV	0.148	2.813	0.005	0.019	Supported
H2a	PEV → CS	0.616	18.044	0.000	0.613	Supported
H2b	PEV → HA	0.656	20.301	0.000	0.754	Supported
H3a	CS → CL	0.369	6.085	0.000	0.148	Supported
H3b	HA → CL	0.413	6.653	0.000	0.186	Supported
Source(s): Developed by authors						

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